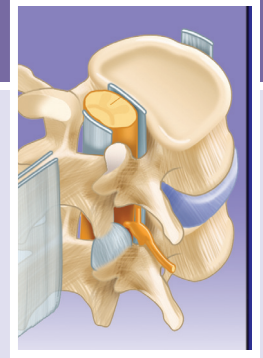


Traumatic Injuries of the Cervical Spine in the Athlete

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This chapter provides an evidence-based approach to history and physical examination, initial stabilization and triage, diagnosis and imaging, treatment, and outcomes for cervical spine injuries in athletes. Cervical spine injury ranges from neuropraxias, to herniated discs, to complete spinal cord injury. The latter is rare but devastating and warrants a high index of suspicion and familiarity with spine immobilization protocols when suspected. On-field catastrophic neurologic injury has decreased over the past 30 years¹; nevertheless, providers must maintain a high index of suspicion, especially in cases of high-velocity axial loading, which is the most common implicated mechanism of injury.² This information is important for team physicians and trainers.

HISTORY

Football, diving, rugby,^{3,4} and other sports constitute the second most common cause (10%) of catastrophic cervical spine injuries (10,000 annually), after motor vehicle collision (MVC).^{5,6} Football is the most researched of these, with 7 decades of Annual Survey of Football Injury Research (ASFIR) data culminating in a reduction of traumatic quadriplegia incidence from 10 to 1 per 100,000 football participants since outlawing spear tackling in 1976.⁷

First responders to an on-field injury should perform primary and secondary surveys, immobilizing the head while ruling out spine injury and assessing airway, breathing, and circulation. Cardiac defibrillation, neurologic exam, equipment removal, and transport are then performed.^{8,9} The neurologic exam should assess level of consciousness, sensation, and strength.

Immobilization in a cervical collar and rigid spine board is an established historical precedent for transport.¹⁰ New recommendations emphasize limited spinal manipulation.¹¹ Spine board immobilization is recommended for palpable step-off or spine tenderness, focal neurologic deficit, altered consciousness, or distracting injury.¹² Medication list, allergies, and time of latest oral intake should be communicated.

PHYSICAL EXAMINATION/ON-FIELD EVALUATION

On-field response requires preparation, planning, equipment, and an organized chain of command. First priorities include assessing scene safety and airway patency. Maintain in-line cervical immobilization while assessing airway, breathing, circulation (ABCs). Note airway obstructions, tracheal deviation, and subcutaneous

emphysema. Check pulses, and apply pressure over active bleeding. Measure heart rate and blood pressure. Assign level of consciousness (alert, responsive to verbal vs. painful stimuli, unresponsive) or Glasgow Coma Scale. Remove clothing to visualize surface anatomy, noting any deformity, crepitus, abrasions, penetrating trauma, burns, tenderness, lacerations, and swelling.

Spine boarding and equipment removal should be decided upon prior to transport.¹³ Some studies support leaving equipment in place until after transport.^{14,15} Even perfect execution engenders a risk of injury.¹⁶ If the helmet or shoulder pads need to be removed, both should be removed to avoid flexion/extension of the cervical spine. Jaw thrust and not head tilt is the appropriate airway access maneuver.¹⁷ Log rolling remains the appropriate technique for prone athletes, to position the spine board beneath them, but fewer cases of iatrogenic neurodeterioration have been documented with newer techniques.¹⁸

To remove equipment, the team leader stands over the head and stabilizes the C-spine while another provider straddles the patient and cuts down the jersey/shoulder pads and under-arm straps.¹⁹ This also facilitates defibrillator pad placement.²⁰ Roles are then reversed, holding C-spine from below while two providers pull off the helmet. If three trained providers are not present, facemask removal is an alternative.²¹ If quick-release mechanisms are not built-in, cordless screwdrivers or heavy shears may prove instrumental in making bilateral cuts or removing the two screws in the plastic clip affixing the facemask.²² Chin strap and cheek pad removal facilitate helmet removal while towels are temporarily placed beneath the head to prevent C-spine extension.²³ Shoulder pads can then slide over the head allowing application of a rigid cervical collar.

The sports medicine team may escalate care via emergency medical transport services and communicate injury mechanism, developments, and interventions during on-field management. Intracranial pressure should be monitored by serial neurologic examination.¹³ Acuity and resource availability should factor into destination selection for transport. Computed tomography (CT), magnetic resonance imaging (MRI), and spine surgeon availability are important destination determinants. Advanced verbal notice to the receiving facility is helpful.

IMAGING

Cervical spine x-rays are important unless the patient is pain-free throughout a full range of motion (ROM), is nontender to

Abstract

This chapter provides an evidence-based approach to sideline medicine, history and physical examination, initial stabilization and triage, diagnosis and imaging, treatment and outcomes, and clearance for play associated with head and neck injuries in athletes. Head and neck injury ranges from sports-related concussion (SRC) to complete spinal cord injury. The latter is rare but devastating and warrants a high index of suspicion and familiarity with spine immobilization protocols whenever head or neck injury is suspected.

Keywords

cervical
stinger
stenosis
transient quadriplegia
spear tackler's spine
strain
spinal cord injury

palpation, is alert and oriented with no distracting injuries or inebriation, and has no neurodeficits.²⁴ With 30,000 patients, the National Emergency X-Ray Utilization Study trial demonstrated 99.8% negative predictive value and 99% sensitivity using these criteria, which reduces unnecessary radiation. Advanced age or high-energy mechanism merit cervical spine x-rays including anteroposterior (AP), lateral, and odontoid views, adding swimmer's view if C7 to T1 is inadequately visualized. Unremarkable radiographic and clinical exam may be followed by flexion/extension views to rule out dynamic instability.

Regular postinjury reevaluation for neurologic weakness or signs of intracranial pressure allows prompt acquisition of advanced imaging.²⁵ This helps to rule out skull fracture, subdural or epidural hematoma, hemorrhage, or edema. However, concussion has no structural findings on imaging.²⁶

Three-dimensional (3D) reconstruction with fine cuts in sagittal, axial, and coronal planes is indicated if clinical exam or x-rays are abnormal. CT scan rapidly identifies injury even without equipment removal,²⁷ with 90% to 100% sensitivity and specificity,²⁸ respectively. Speed, power, subtlety, and cervicothoracic visibility have led CT to replace plain films as first line imaging. Maintenance of spinal precautions during transfers is imperative. C-collar should remain in place until clearance of the C-spine.

MRI is indicated²⁹ for neurodeficits and soft tissue characterization of posterior ligamentous complex (PLC), herniated nucleus pulposus (HNP), and spinal cord injury or if spinal cord injury without radiographic abnormality (SCIWORA) is suspected. Neurodeficits or radiographic injury merit spine surgery consultation.

CT head should be acquired in cases of open, depressed, or basilar skull fracture, persistent headache or vomiting, high-energy mechanism, anterograde amnesia, or evidence of supraclavicular trauma.³⁰ Delayed electromyography provides baseline and prognostic information about preganglionic versus postganglionic brachial plexopathies, with isolated motor (poor) versus combined motor and sensory (favorable) potential loss, respectively.³¹

DECISION-MAKING PRINCIPLES OF DIAGNOSIS AND TREATMENT

Cervical sprain/strain is a diagnosis of exclusion, characterized by unremarkable plain radiographs in the context of mild neck tenderness and stiffness devoid of paresthesias or neurodeficit. It can be monitored closely and managed expectantly with rest and a low threshold for CT scan. Stingers or burners are transient upper extremity weakness (often C5 or 6 root) often accompanied by paresthesias with negative imaging findings and approximately 50% likelihood of lifetime recurrence.³² They result from traction injuries to nerve roots³³ or the brachial plexus.³⁴ Another neuropraxia, burning hands syndrome, is a central cord variant without permanent deficit characterized by bilateral weakness and/or dysesthesias of the hands corresponding to spinothalamic/corticospinal tract injury without imaging findings.³⁵ This too is self-limiting. Transient quadriplegia (also known as spinal cord concussion) is an acute, traumatic, self-limited total paralysis lasting minutes to hours with an incidence of 7 per 10,000 football players. It may result from a pincer mechanism³⁶ without predisposition to recurrent neurologic symptoms.³⁷

Herniated cervical intervertebral disc or HNP can occasionally be an acute traumatic entity with central and/or neuroforaminal stenosis experienced as axial/periscapular neck pain with upper extremity dermatomal myeloradiculopathy, paresthesias, weakness, gait ataxia, or loss of manual dexterity, and sometimes bowel/bladder dysfunction with quadriplegia. Neurologic deficit is an indication for urgent MRI and surgical decompression.² Surgery includes anterior cervical discectomy and fusion (ACDF) for central compression or posterior foraminotomy for foraminal HNP.

Spinous and transverse process and vertebral body fractures with less than 20% height loss often are stable nonoperative injuries, demonstrating less than 3.5 mm or 11 degrees of translational or angular instability, respectively, on flexion/extension radiographs.³⁸ Atlantoaxial instability is often the result of transverse alar ligament (TAL) injury, which manifests in atlantodens interval (ADI) widening beyond 3 mm on midsagittal CT.³⁹ TAL injury in Jefferson C1 ring fractures⁴⁰ are unstable when the combined bilateral overhang of C1 over C2 lateral masses on odontoid view or coronal CT exceeds 7 mm. Atlantoaxial instability is treated with C1/2 fusion.

Anderson and D'Alonzo classify C2 dens fractures⁴¹ into alar ligament tip avulsions (type 1), treated in C-collar; watershed area fracture through dens waist (type 2), treated with fusion/osteosynthesis in cases of risk factors for nonunion; and base of dens fracture through cancellous body of axis (type 3), which go on to union with collar immobilization.

Traumatic spondylolisthesis of C2 (hangman's fractures) are classified by Levine and Edwards⁴² into minimally displaced (type 1), treated with rigid orthosis; angulated (type 2A), treated with halo vest; angulated and translated (type 2B), treated with traction and halo vest; and bilateral pars fractures with subaxial facet dislocation (type 3), treated with posterior arthrodesis.

Single column compression fractures of the subaxial cervical spine⁴³ can be treated in a C-collar. Unstable³⁸ two-column burst injuries are treated with anterior or posterior arthrodesis and possible corpectomy. Anterior-inferior teardrop fractures require CT evaluation of coronal and sagittal plane injury and surgical stabilization due to a high incidence of quadriplegia.⁴⁴

Hyperflexion injuries can cause unshingling and unilateral or bilateral facet dislocation visible on x-ray (25% or 50% intersegmental subluxation, respectively) or CT imaging and should be reduced using cervical tongs and traction in the awake and oriented patient. Any change in neurologic status or failed reduction prompts MRI to rule out HNP prior to emergent open reduction.⁴⁵ Bilateral versus unilateral facet dislocation has a much higher rate of quadriplegia.

Cervical stenosis is defined as canal AP diameter less than 14 mm on lateral x-ray or Torg ratio (AP diameter of canal divided by vertebral body) less than 0.8⁴⁶ and may predispose to spinal cord injury. Complete spinal cord injury has an incidence of 0.33 in high school and 1.33 in college athletes, with 11,000 cases and \$9.7 billion cost annually. It is a devastating and irreversible injury encountered most often in professional football players making a defensive tackle.⁴⁷ Emergent decompression and stabilization is indicated.

Central cord syndrome is an incomplete hyperextension injury in elderly congenitally narrowed canals producing upper and distal more than lower and proximal musculature weakness. This is the most common and least devastating incomplete injury.⁴⁸ Brown-Séquard is a penetrating trauma-induced cord hemisection syndrome characterized by ipsilateral corticospinal tract motor paralysis and contralateral pain and temperature sensation disruption. Its prognosis is also favorable. Anterior cord syndrome, the most devastating incomplete injury, can be the result of ischemic or compressive injury to the anterior two-thirds of the cord from anterior spinal artery stroke or HNP/hyperflexion disrupting all but the dorsal columns. Rarely, one sees isolated posterior artery stroke of the dorsal columns and isolated loss of proprioceptive and vibratory sense in posterior cord syndrome.

Sprains and strains are treated with activity modification and cautious gameplay to tolerance until resolution of pain, stiffness, and tenderness. Neuropraxias are treated with withdrawal from game play and rest until resolution of all paresthesias, weakness, and pain with ROM.⁴⁹ Recurrent neuropraxia warrants equipment modification and physical therapy for shoulder girdle strengthening. Thereafter gameplay may resume. Symptomatic HNP without weakness or deficit can be treated nonoperatively. Single-level ACDF procedure is compatible with return to competitive sports,³ and return to play (RTP) can be gradually initiated after radiographic evidence of fusion. Fractures and unstable injuries of the cervical spine should be evaluated by a spine surgeon to determine treatment options.

RETURN TO PLAY

Spine precautions should be maintained until full compliance with neurologic exam is possible. C-collar necessity can be reevaluated 1 to 2 weeks after injury in clinic with flexion/extension radiographs. RTP requires physical/cognitive rest until asymptomatic prior to return to school⁵⁰ and neurologic and musculoskeletal exam and concussion reassessment prior to clearance.⁵¹ Twenty-four-hour cycles of light aerobic activity, mildly elevated heart rate, sport-specific and noncontact drills and medical clearance prior to full-contact practice are recommended.⁵²

After cervical sprain, athletes may return to sport when full painless ROM has been reestablished without tenderness to palpation. After neuropraxic injury, strength, as well as pain-free ROM of the C-spine and extremities and complete resolution of paresthesias, numbness, and tingling, are necessary prior to returning to competitive play. Players with stingers that resolve completely or transient quadriplegia that lasts less than 24 hours without any residual deficits may also RTP without contraindication, provided this was the patient's first episode. Three episodes of stingers, prolonged symptoms, or transient quadriplegia lasting more than 24 hours are considered relative contraindications to return to competitive contact sport. Absolute contraindications include multiple episodes of transient quadriplegia, clinical or MRI evidence of cervical myelopathy, and cervical pain or neurologic deficits.

After an athlete has sustained a neuropraxia, he or she may RTP if the previous criteria are met and the Torg ratio is greater



Fig. 128.1 Congenital cervical stenosis. Torg ratio: Anteroposterior (AP) diameter of spinal canal divided by AP diameter of corresponding vertebral body: greater than 0.8 normal, less than 0.8 congenital stenosis, and increased risk of neurologic injury.

than 0.8, defined as the ratio of the AP diameter of the canal divided by that of the vertebral body. Torg ratio less than 0.8 is consistent with a diagnosis of congenital cervical stenosis (Fig. 128.1), which places the athlete at increased risk of neurologic injury. Imaging evidence of intervertebral disk herniation, degenerative change, or cord deformation on MRI in this context is considered a relative contraindication for RTP. If there is MRI evidence of cord injury, defect, edema, or myelomalacia in the context of prior neuropraxic episode the athlete should be absolutely contraindicated from return to competitive contact sports. Likewise, evidence of ligamentous instability, prolonged symptoms greater than 36 hours, or multiple neuropraxic episodes should constitute an absolute contraindication. Numerous reports of professional-level elite contact and power sport athletes safely returning to same level of play after single level ACDF surgery have been published. Therefore evidence suggests that short-segment fixation in the cervical spine is not a contraindication for competitive sport. Individualized assessment of severity of pathology and surgical complexity will determine the ultimate competitive potential and timeline for RTP in these cases.

Spear tackler's spine is another absolute contraindication to competitive sport that risks axially loading a vulnerable cervical spine. A subset of football players demonstrate this condition, which includes findings of congenital stenosis, loss of physiologic lordosis (Fig. 128.2), posttraumatic changes on imaging of the cervical spine, and/or behavioral history of spear tackling techniques during gameplay. Players demonstrating these characteristics should be barred from competitive football or other contact sports. Coaching regarding proper tackling technique (Fig. 128.3)

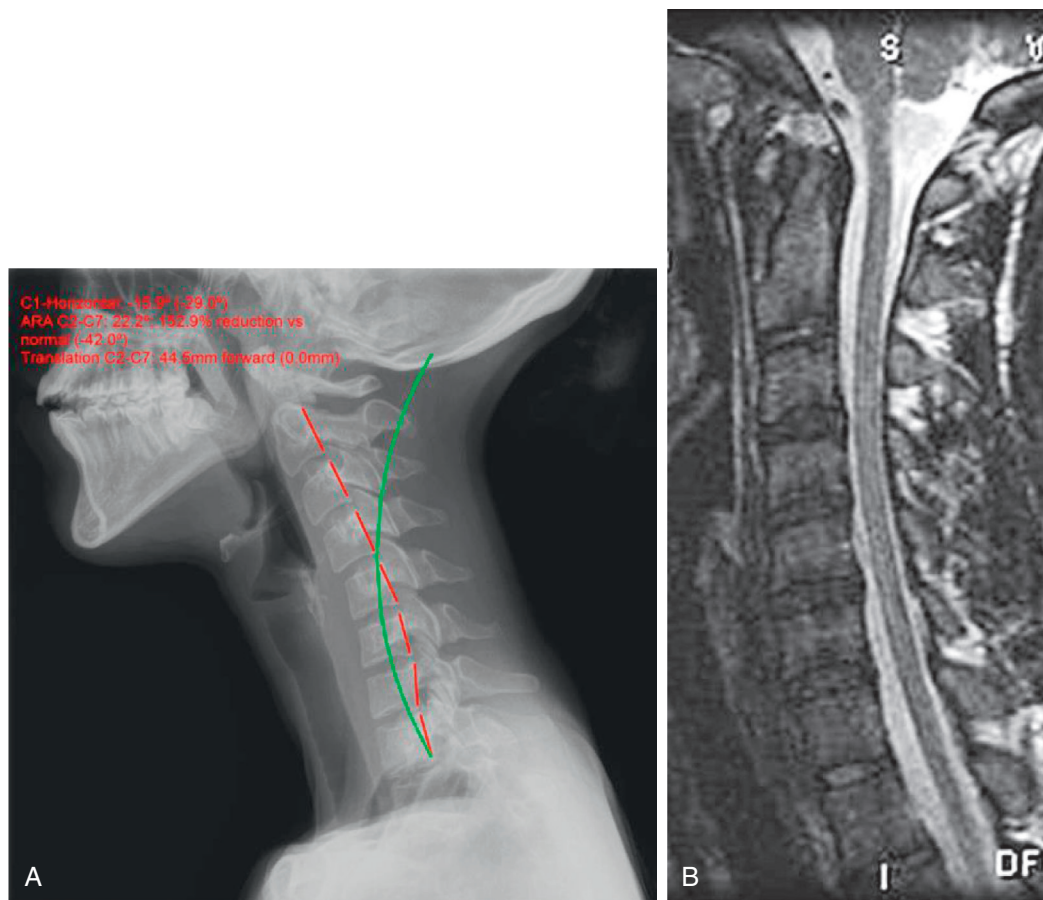


Fig. 128.2 Spear tackler's spine. (A) Loss of the normal lordosis of the cervical spine (standing neutral x-ray), (B) congenital stenosis of the central canal (recumbent sagittal magnetic resonance imaging), posttraumatic changes resembling spondylosis of multilevel disc bulge and ligamentous hypertrophy.



Fig. 128.3 Proper tackling technique involves a low center of gravity, directing the shoulder at the ball-carrier's beltline. Crucially, the tackler keeps his/her head up at all times and wraps up the ball-carrier's legs, so he/she falls to his or her side, taking both players to the ground together.

is crucial to avoid catastrophic neurologic injury from axially loading the cervical spine during spear tackling.

CONCLUSIONS

Physician, trainer, emergency medical services (EMS), and others are part of a team-based approach to sports medicine.⁵³ Team preparation should be performed annually and before each practice/game. Communication is critical between trainers and teachers for customized student-athlete care.⁵⁴ This is also important for post-return to sport monitoring.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

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